

Deep Learning-Based Detection of Mangrove Gain and Loss in Aringay La Union

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Situation Analysis

WHY MANGROVES MATTER

- Coastal defense & biodiversity habitat
- Carbon storage & environmental balance

MANGROVES IN DECLINE

- Global cover significantly decreased; loss acute in Southeast Asia (Worthington et al., 2020; FAO, 2023; Lomeo & Singh, 2022)
- Philippines: historically 400,000–500,000 ha; considerable reductions since (Pillodar et al., 2023; Sasi et al., 2025)
- Drivers vary by region: aquaculture, reclamation, land-use change (Primavera, 2000)

PROVEN TOOLS, FREE DATA

- Deep learning + remote sensing now standard for environmental monitoring; CNNs lead (Zhang et al., 2025)
- Sentinel-2: freely available, recurrent multispectral imaging (ESA, 2024)

What Existing Work Does Not Cover

- ▶ Coarse global & national products — Global Mangrove Watch & the 2023 Philippine Mangrove Map use low resolutions and generalized methods → may miss localized transitions, fragmentations, narrow coastlines



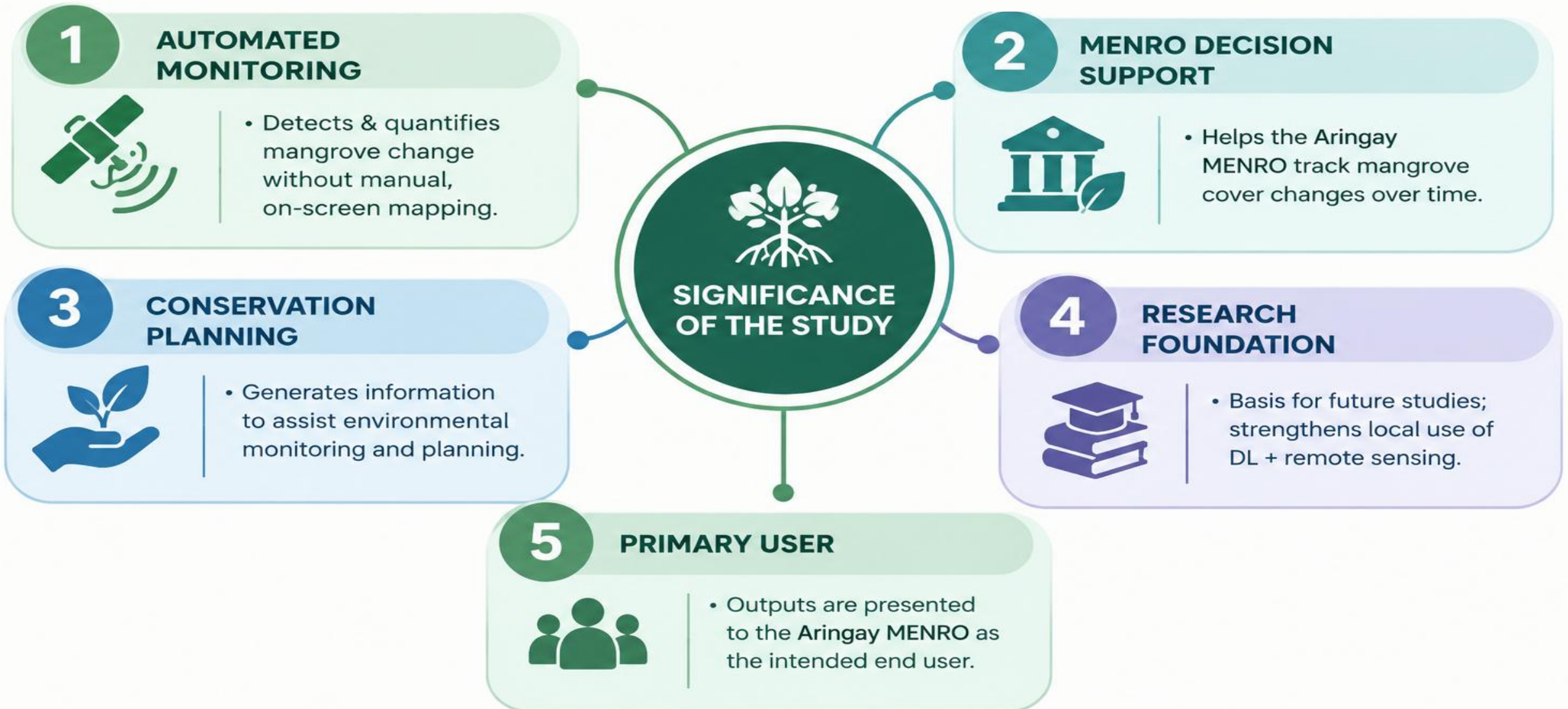
- ▶ No localized dataset — U-Net + Sentinel-2 proven in Myanmar, Florida, and Bohol; no localized dataset exists for Barangay Dulao

The solution: a localized mangrove dataset + a combined gain-and-loss detection approach for Barangay Dulao, Aringay, La Union.



National Policy Alignment — Supports ecosystem accounting under the Philippine Ecosystem and Natural Capital Accounting System (PENCAS) Act, RA 11995, providing an objective monitoring tool for local environmental governance.

Significance of the Study



Statement of Objectives

General objective: To develop an image processing system that automatically detects and quantifies mangrove cover change over time in Aringay, La Union, using satellite imagery and deep learning-based segmentation.

SPECIFIC OBJECTIVES

- 1 Collect a localized mangrove dataset
- 2 Train the mangrove gain-and-loss detection model using U-Net
- 3 Evaluate model performance in terms of F1-score, Intersection over Union (IoU), and mean Average Precision (mAP)

TIME and PLACE

SATELLITE IMAGE

2019



2024



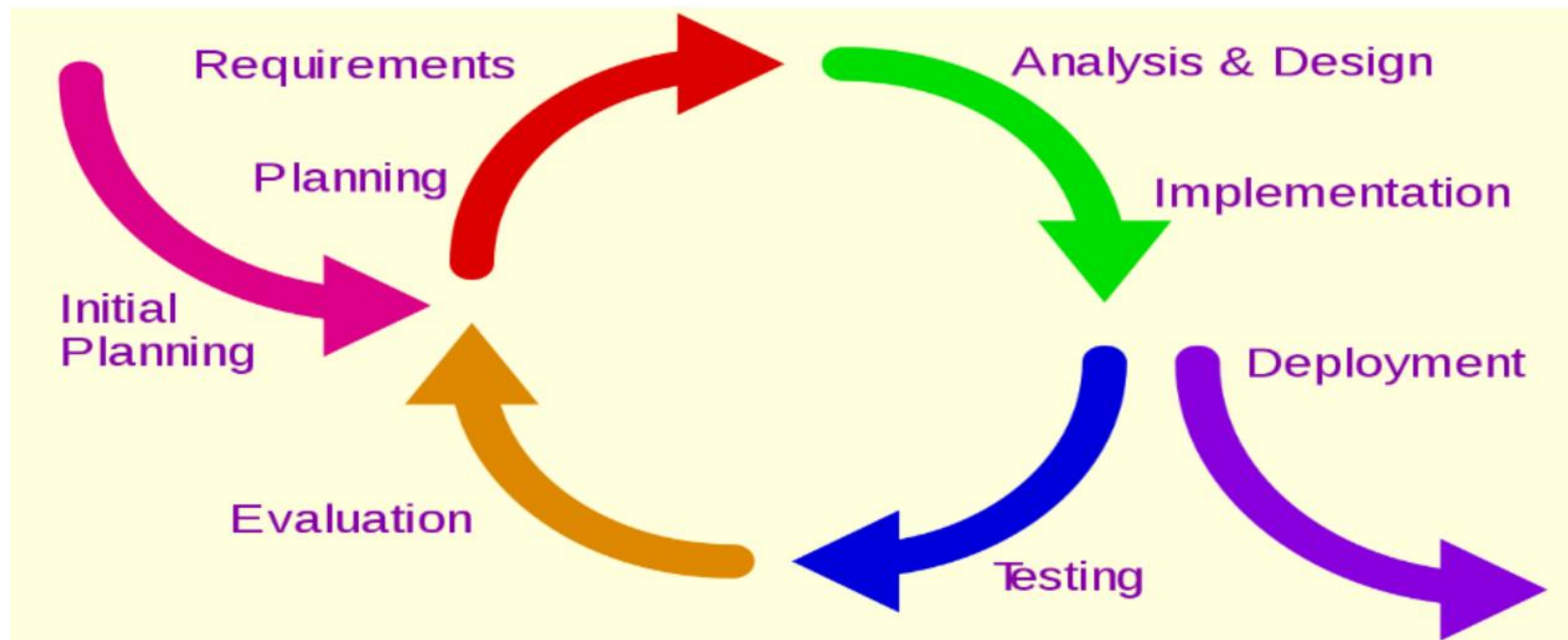
COMPARISON

SATELLITE IMAGE

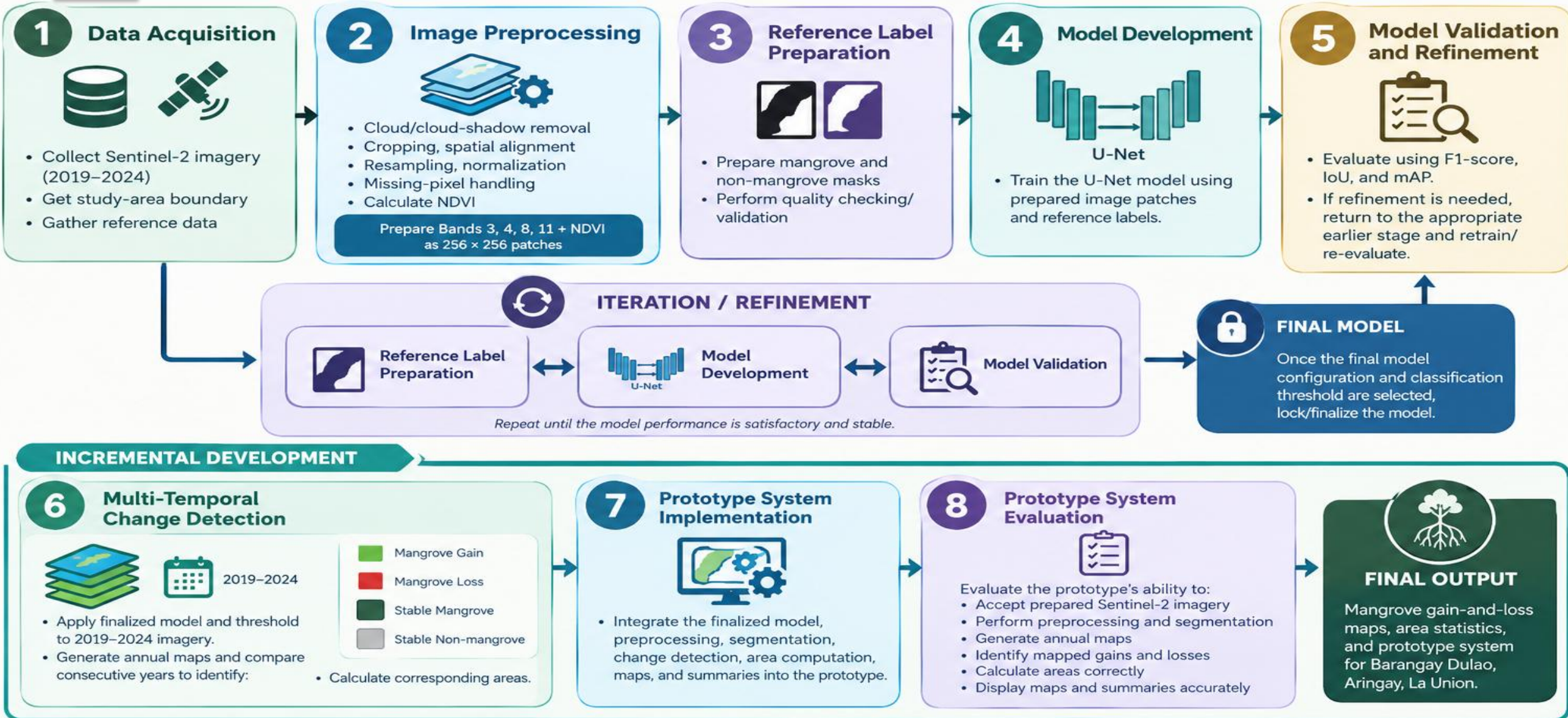


Research Design

- Approach: Hybrid Experimental–Descriptive Research Design
- Framework: Iterative and Incremental Development (IID)
- Purpose: Guides development through repeated cycles of building, testing, and refinement (Larman & Basili, 2003; Ibrahim et al., 2020).



Research Workflow — Iterative & Incremental Development



Dataset Collection & Preparation

PRIMARY DATA — SENTINEL-2 IMAGERY

- Barangay Dulao mangrove area 2019–2024
- Bands: B3 Green, B4 Red, B8 NIR, B11, SWIR
- $NDVI = (B8 - B4) / (B8 + B4)$ — explicit vegetation-health signal
- Selection: full coverage · minimal cloud · low shading

PREPARATION

- Cloud/shadow removal · crop to boundary · grid alignment · normalization · missing-pixel handling
- B11 resampled 20 m → 10 m (bilinear) to unify the stack
- 256 × 256 patches · 5-channel input · pixel-aligned masks

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DATA LABELING & MENRO VALIDATION

BINARY MASK PREPARATION

1

=

MANGROVE

0

= **NON-**

MANGROVE

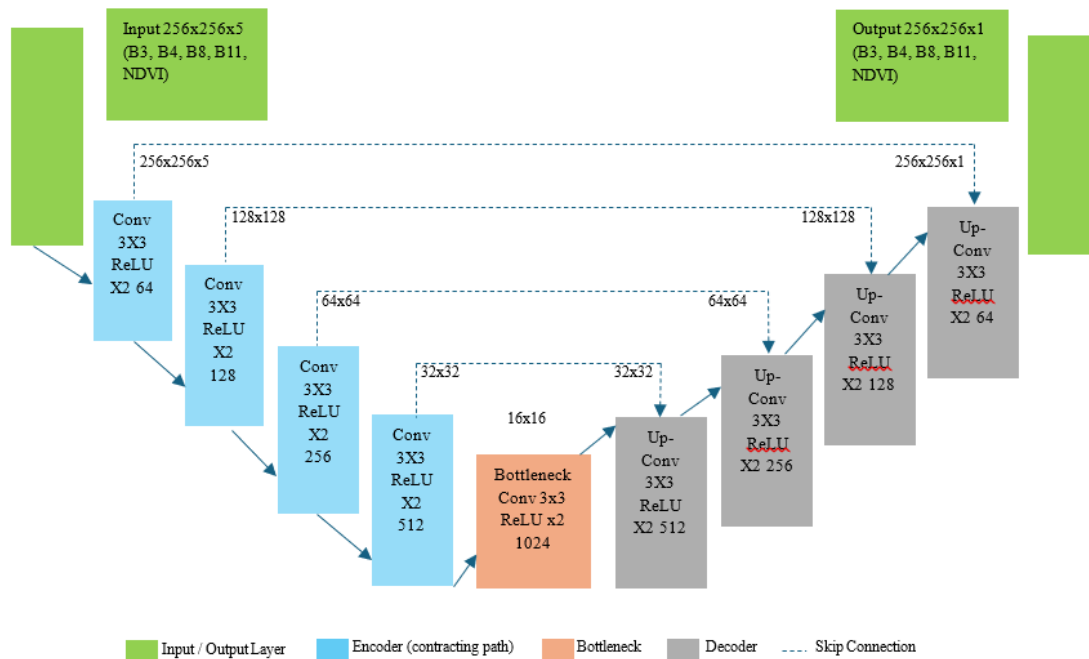
Every pixel in the reference mask is labeled using available mangrove datasets, local records, and high-resolution imagery — assigned a value of 1 if it is mangrove, or 0 if it is not.

INDEPENDENT MENRO VALIDATION

The researchers who prepare the initial labels are the same people who train and evaluate the model.

Left unchecked, this is a conflict of interest — the model could simply be graded against its own creators' assumptions.

U-Net Architecture



WHY U-NET FITS THIS TASK

- ▶ Goal is pixel-level classification, not whole-image classification
- ▶ Skip connections preserve mangrove boundaries & small/fragmented patches
- ▶ Proven in comparable settings; lower data & compute than transformers (Ronneberger et al., 2015)

TRAINING CONFIGURATION

- ▶ Adam optimizer · learning rate 0.001 · binary cross-entropy loss
- ▶ Batch size, epochs, augmentation, class-imbalance handling, stopping criteria recorded
- ▶ Training vs. validation loss monitored for overfitting/underfitting

Model Validation and Refinement

F1-SCORE

$$2TP / (2TP + FP + FN)$$

What it measures?

Balance between correctly identified and missed mangrove pixels.

IoU

$$TP / (TP + FP + FN)$$

What it measures?

Overlap between predicted and reference mangrove areas.

mAP

$$AP = \sum (R_n - R_{n-1}) \times P_n$$

What it measures?

Model performance across different classification thresholds.

Change Detection: Mangrove Gain & Loss

Locked model applied uniformly to all years → annual binary maps 2019–2024

2019–2020	2020–2021	2021–2022
2022–2023	2023–2024	Overall 2019–2024

Area statistics: 10 m grid → 1 pixel = 100 m²

$$\text{Area(ha)} = N \times 100 \text{ m}^2 / 10,000 \text{ m}^2 / \text{ha}$$

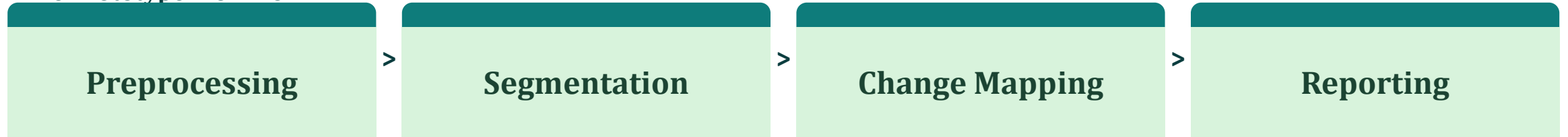
$$\text{Area(ha)} = N / 100$$

Where *N* represent the number of classified pixel

EARLIER YEAR	LATER YEAR	TRANSITION CATEGORIES
Non-mangrove	Mangrove	Mapped mangrove gain
Mangrove	Non-mangrove	Mapped mangrove loss
Mangrove	Mangrove	Stable mangrove
Non-mangrove	Non-mangrove	Stable non-mangrove

Prototype Implementation & Evaluation

The Prototype Workflow



What It Does

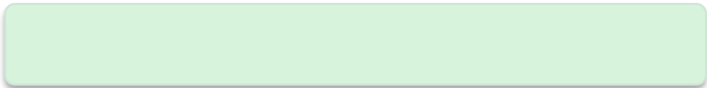
- Accepts prepared Sentinel-2 imagery
- Executes preprocessing and segmentation automatically
- Generates annual mangrove-cover maps
- Identifies mapped gain and loss between years
- Calculates corresponding area in hectares
- Displays maps, tables, and summary information to the user

Functional Evaluation (Phase 8)

Rather than another model-training or model-selection stage, this phase verifies the system works as intended — confirming each step above executes correctly, with area calculations independently re-verified from the classified pixel counts.

Practical Tool Delivery

The completed prototype presents its outputs — maps, tables, and area statistics — directly to Aringay MENRO, delivering a practical, working tool rather than a theoretical model alone.



THANK YOU

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